

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-ab6befa5da96aae45.jsonl`  
- SHA-256: `bd00a8f15b435c05c5ba6607efa5c384e5f430ab5a6ffdb35b07d4265359aba3`  
- Source modified: `2026-06-16T03:20:04+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:19:26.158Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://pmc.ncbi.nlm.nih.gov/articles/PMC11798530/>

****Title:**** Pseudo Label Refining (PLR) for Semi-Supervised Temporal Action Localization (PLOS One, Feb 2025)

****Found via:**** Small-data / weak / timestamp-supervised TAL + active learning search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:19:27.964Z)

I'll fetch the source page and extract the relevant claims.

3. user (2026-06-16T03:19:42.074Z)

Key Technical Claims: Pseudo Label Refining (PLR) for Semi-Supervised Temporal Action

(1) Method and Problem

****What it solves:**** The paper addresses noise in pseudo-labels during semi-supervised temporal

action localization (SS-TAL). "The noise within pseudo-labels directly affects the effectiveness of semi-supervised learning" as pseudo-labels contain classification errors and boundary offsets.

The PLR method consists of three components:

- **Pseudo-Label Self-Refinement (PLSR):** Adapts bounding box refinement from object detection to 1D temporal boundaries using linear interpolation, 1D convolution, and regression to refine action intervals
- **Boundary Synthesis (BS):** "leveraging information from surrounding non-maximum predictions" through random boundary disturbance, Gaussian noise addition, and confidence-weighted averaging across K iterations
- **Adaptive Weight Learning (AWL):** Assigns dynamic weights based on mean information entropy, "prioritizing high-confidence labels and gradually incorporating lower-confidence ones"

(2) Supervision Setting

Four label rates tested: "10%, 20%, 40%, and 60%" of samples randomly selected per action category as labeled data, with remaining samples unlabeled.

(3) Datasets and Results

Datasets: THUMOS14 and ActivityNet v1.3

Key mAP improvements at 60% label rate:

- THUMOS14 with ActionFormer: "61.2%" (vs. NPL baseline "59.0%")
- ActivityNet v1.3 with ActionFormer: "36.0%" (vs. NPL baseline "35.8%")
- THUMOS14 with BMN: "42.9%" (vs. NPL baseline "42.2%")

"The experimental results show that the method trained with PLR outperforms the previous state-of-the-art methods by up to 3.5% mAP"

(4) Pseudo-Label Noise Handling

Noise sources identified: "classification error" and "boundary offset" in pseudo-labels.

Thresholding: "All pseudo labels are filtered based on a class-confidence threshold of 0.2."

Quality measurement: Mean information entropy (mIE) used to measure pseudo-label credibility: "When the output corresponding to the classification is more evenly distributed, the mIE is higher, and the corresponding pseudo-labels are less reliable."

(5) Publication Details

Date: "2025 Feb 5"

Venue: PLoS One, Volume 20, Issue 2, article e0318418

DOI: 10.1371/journal.pone.0318418

4. assistant (2026-06-16T03:19:45.090Z)

I have the extracted content. This is a primary research paper published in PLoS One (peer-reviewed). Let me assess and structure the claims relevant to the research question (specifically Dimension 3: small-data strategies, semi-supervised TAL, pseudo-labeling).

5. user (2026-06-16T03:19:56.612Z)

Structured output provided successfully

6. assistant (2026-06-16T03:20:04.590Z)

Extracted and returned the structured findings. Summary of what I found:

Source: PLoS One peer-reviewed primary research (Feb 5, 2025), DOI 10.1371/journal.pone.0318418 ? rated `primary`.

****Relevance to the research question (Dimension 3 ? small-data / semi-supervised TAL):****
Moderate. The paper's PLR method directly addresses pseudo-label noise in semi-supervised TAL, with two transferable ideas: (1) boundary refinement of noisy pseudo-labels via 1D regression ? directly relevant to the over-merge/boundary problem, and (2) entropy-weighted curriculum incorporation of pseudo-labels ? relevant to label-noise handling if Gemini is used as an inference-time pseudo-labeler.

****Key caveats flagged for the report:****

- The gains are modest in absolute terms (2.2 pt mAP at 60% labels on THUMOS14/ActionFormer; "up to 3.5%" headline).
- The label rates tested (10?60%) are far higher than the target system's extreme low-label regime (~6?16 videos), so transfer to that scale is ****unverified****.
- The method is built on ActionFormer/BMN over generic action datasets (THUMOS14, ActivityNet), not 1-D shuttle-trajectory + cue signals ? applicability is by analogy, not demonstrated.

All claims include direct quotes and are rated central/supporting/tangential per the schema.